Extended sign-kill for $\widehat{\lambda} = (4,3)$: four-piece E^+_{n-1} decomposition with two vanishing branches

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Abstract

Let $\lambda = (4,3,1^q) \vdash n = q+7$ with $q \geq 5$, $\ell = q+2 = n-5$, and $B = B^{(\lambda)}_{\ell+1} V_\lambda = R'_{\ell+1} R'_{\ell+2} R'_{\ell+3} R'_{\ell+4} R'_{\ell+5}\, V_\lambda$ (five R'-blocks). We prove the "$\subseteq$" direction of the extended sign-kill conjecture for this family at the predicted threshold $\tau = q+3-\widehat{\lambda}_1-\widehat{\lambda}_2 = q-4$:

$$B \;\subseteq\; \bigcap_{j=1}^{q-4} E^-_j\,|_{V_\lambda}\,.$$

At $q \in \{0,1,2,3,4\}$ the predicted threshold is non-positive, so the containment is vacuous.

The shape $\widehat{\lambda} = (4,3)$ is the *first* case in the extended sign-kill programme where the inner E^+_{n-1} decomposition has four pieces (three from corner-pairs plus one same-row hook), and where *two* of those pieces vanish in the pullback (μ_{12} via the $\widehat{\lambda} = (3,2)$ recursive input, μ_h via the hook column-1 theorem), leaving two non-zero contributions (μ_{13} via the $\widehat{\lambda} = (3,3)$ recursive input, μ_{23} via the $\widehat{\lambda} = (4,2)$ recursive input). The dimensions match: $\dim B = f^{(3,2)} = 5 = f^{(2,2)} + f^{(3,1)} = \dim B^{(\mu_{13})}_{\ell+1} + \dim B^{(\mu_{23})}_{\ell+1}$.

This is the first fully-recursive three-branch case where every contributing inner shape is a non-hook treated by a prior extended sign-kill theorem in the May-14 series.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \ldots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$.

For $1 \leq j \leq n-1$, set $S_j := T_j + 1$. Write

$$E^+_j \;:=\; \ker(T_j - q)\,|_{V_\lambda} = \;\mathrm{Im}(S_j\,|_{V_\lambda}), \qquad E^-_j \;:=\; \ker(S_j)\,|_{V_\lambda}\,.$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega \;=\; \Omega^{(\lambda)} \;:=\; R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \qquad B \;:=\; \Omega \cdot V_\lambda \;=\; B^{(\lambda)}_{\ell+1} V_\lambda.$$

For $\lambda = (4,3,1^q)$, $n = q+7$ and $\ell = q+2$, so $\Omega = R'_{q+3} R'_{q+4} R'_{q+5} R'_{q+6} R'_{q+7} = R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n$ has *five* R'-blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda^{(\geq 2)}} = f^{(3,2)} = 5$ in the stable regime $q \geq 2$.

Removable corners of λ. The removable corners are

$$c_1 = (1,4), \qquad c_2 = (2,3), \qquad c_3 = (q+2,1).$$

λ has two length-preserving corners (c_1, c_2) and one column-1 bottom corner (c_3), an $r = 2$ shape.

Identity prefix and seminormal support. For an SYT T of any partition μ, the *identity prefix length* is

$$p(T) \;:=\; \max\{\, P \geq 0 : T(i,1) = i \text{ for all } 1 \leq i \leq P \,\}.$$

For $W \subseteq V_\mu$, $\operatorname{supp}(W) := \{\, T \in \mathrm{SYT}(\mu) : \exists w \in W,\ [v_T]\, w \neq 0 \,\}$.

2 Input from prior papers

We collect the prior extended sign-kill theorems used as recursive inputs.

Proposition 1 (Support rule, May-14 fourth pass)**.** *Let $W \subseteq V_\nu$ be any subspace, $1 \leq j \leq |\nu| - 1$. If every $T \in \operatorname{supp}(W)$ has j and $j+1$ in the same column of T, then $W \subseteq E_j^-|_{V_\nu}$.*

Remark 2. If every $T \in \operatorname{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P-1$ the entries $j, j+1$ sit at column-1 rows $j, j+1$, sharing column 1. By Proposition 1, $W \subseteq E_j^-$ for $1 \leq j \leq P-1$.

Theorem 3 (Extended sign-kill at $\widehat{\lambda} = (3,2)$, May-14 fifth pass)**.** *For $\lambda' = (3,2,1^{q'}) \vdash n' = q'+5$ with $q' \geq 3$, every $T' \in \operatorname{supp}\big(B^{(\lambda')}_{q'+3} V_{\lambda'}\big)$ satisfies $p(T') \geq n' - 6 = q' - 1$. Consequently, $B^{(\lambda')}_{q'+3} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q'-2$, and in particular $B^{(\lambda')}_{q'+3} \subseteq E_1^-$ for $q' \geq 3$.*

Theorem 4 (Extended sign-kill at $\widehat{\lambda} = (4,2)$, May-14 sixth pass)**.** *For $\lambda' = (4,2,1^{q'}) \vdash n' = q'+6$ with $q' \geq 4$, every $T' \in \operatorname{supp}\big(B^{(\lambda')}_{q'+3} V_{\lambda'}\big)$ satisfies $p(T') \geq n' - 8 = q' - 2$. Consequently, $B^{(\lambda')}_{q'+3} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q'-3$, and in particular $B^{(\lambda')}_{q'+3} \subseteq E_1^-$ for $q' \geq 4$.*

Remark 5. Theorem 4 as stated in the sixth-pass paper has a mild gap: the inner $E^+_{n'-1}$ decomposition omits a same-row $+q$-piece at $c_1 = (1,4)$ with inner shape $(2,2,1^{q'})$. The seventh-pass paper (Remark 3.7 there) and the present paper note that this gap is closable by an additional application of the $\widehat{\lambda} = (2,2)$ extended sign-kill theorem (May-14 fourth pass): the missing same-row piece gives an inner subspace inside $E_1^-|_{V_{(2,2,1^{q'})}}$ (the $(2,2)$ theorem at $q'' = q'$), and the rightmost $(T_1{+}1)$ of an extra $R'^{(2,2,1^{q'})}_{|.|-3}$ factor kills it. The conclusion of Theorem 4 (containment direction) is correct as stated; we apply it below.

Theorem 6 (Extended sign-kill at $\widehat{\lambda} = (3,3)$, May-14 seventh pass)**.** *For $\lambda' = (3,3,1^{q'}) \vdash n' = q'+6$ with $q' \geq 4$, every $T' \in \operatorname{supp}\big(B^{(\lambda')}_{q'+3} V_{\lambda'}\big)$ satisfies $p(T') \geq n' - 8 = q' - 2$. Consequently, $B^{(\lambda')}_{q'+3} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q'-3$, and in particular $B^{(\lambda')}_{q'+3} \subseteq E_1^-$ for $q' \geq 4$.*

Theorem 7 (Hook column-1, May-12)**.** *For $\mu = (k, 1^m) \vdash n_0$ with $k \geq 2$ and $m \geq \max(k,2)$, $B^{(\mu)}_{\ell_0+1} V_\mu \subseteq E_1^-|_{V_\mu}$ (where $\ell_0 = \ell(\mu) = m+1$).*

Theorem 8 (Phase A endpoint, May-19)**.** *For any H_q-module V and any $1 \leq j \leq n-1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E^+_{n-1}|_{V_\lambda}$.*

Lemma 9 (Outer-factor preservation, May-14 fifth pass)**.** *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \operatorname{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P+1$, every $T'' \in \operatorname{supp}((T_j{+}1)W)$ also satisfies $p(T'') \geq P$.*

3 The four-piece E^+_{n-1} decomposition for $(4, 3, 1^q)$

The $+q$-eigenspace $E^+_{n-1}|_{V_\lambda}$ admits a clean four-piece decomposition reflecting the corner structure of λ.

2-block pieces. For each pair $\{c_X, c_Y\}$ of removable corners with $c_X \neq c_Y$, placement of $\{n-1, n\}$ at $\{c_X, c_Y\}$ in either order yields a T_{n-1}-2-block in V_λ. The pairs and axial distances:

- $\{c_1, c_2\} = \{(1, 4), (2, 3)\}$, axial distance $|c(1, 4) - c(2, 3)| = |3 - 1| = 2$. Non-degenerate.
- $\{c_1, c_3\} = \{(1, 4), (q + 2, 1)\}$, axial distance $|3 - (1 - (q + 1))| = q + 3 \geq 8$ at $q \geq 5$. Non-degenerate.
- $\{c_2, c_3\} = \{(2, 3), (q + 2, 1)\}$, axial distance $|1 - (1 - (q + 1))| = q + 1 \geq 6$ at $q \geq 5$. Non-degenerate.

The doubly-stripped inner shapes are

$$\mu_{12} = (3, 2, 1^q), \quad \mu_{13} = (3, 3, 1^{q-1}), \quad \mu_{23} = (4, 2, 1^{q-1}),$$

each a valid partition.

Same-row pieces. A same-row $+q$-piece exists for T_{n-1} on V_λ when there is a removable corner c and an SYT of λ with $n-1$ at the cell immediately left of c (same row) and n at c. The candidates:

- At $c_1 = (1, 4)$: would require $T(1, 3) = n - 1$, $T(1, 4) = n$. But the column-3 constraint $T(1, 3) < T(2, 3)$ would force $T(2, 3) > n - 1$, hence $T(2, 3) \geq n$, impossible since n sits at $(1, 4)$. *No same-row piece at c_1.*
- At $c_2 = (2, 3)$: requires $T(2, 2) = n - 1$, $T(2, 3) = n$. The remaining entries $\{1, \dots, n - 2\}$ must fill $\lambda \setminus \{(2, 2), (2, 3)\}$. The cells form the shape $(4, 1, 1^q) = (4, 1^{q+1})$, a valid partition. *Same-row piece at c_2 with inner hook $\mu_h := (4, 1^{q+1})$.*
- At $c_3 = (q + 2, 1)$: no cell to its left. *No same-row piece.*

No other pieces. Same-column pairs at column 1 give -1-eigenvectors of T_{n-1} (not in E^+_{n-1}).

Lemma 10 (Four-piece decomposition of E^+_{n-1}). *Define embeddings $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ for $X \in \{12, 13, 23\}$ on the seminormal basis by*

$$\Phi_X(v_{T'}) \;:=\; v_{T^+_X} + \tau_X \, v_{T^-_X},$$

where T^+_X, T^-_X are the two SYTs of λ obtained from $T' \in \mathrm{SYT}(\mu_X)$ by placing $\{n-1, n\}$ at the pair of cells $\{c_X, c_Y\}$ in the two possible orders, and $\tau_X \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at the corresponding axial distance. Define $\Phi_h : V_{\mu_h} \hookrightarrow V_\lambda$ on the seminormal basis by $\Phi_h(v_{T''}) := v_{T'''}$, where T''' is the unique SYT of λ obtained from $T'' \in \mathrm{SYT}(\mu_h)$ by placing $n - 1$ at $(2, 2)$ and n at $(2, 3)$. Then:

(i) Each Φ_X ($X \in \{12, 13, 23, h\}$) is injective and T_i-equivariant for $1 \leq i \leq n - 3$.

(ii) The four images in V_λ have pairwise disjoint seminormal supports and satisfy

$$R'_n V_\lambda \;=\; E^+_{n-1}|_{V_\lambda} \;=\; \Phi_{12}(V_{\mu_{12}}) \,\oplus\, \Phi_{13}(V_{\mu_{13}}) \,\oplus\, \Phi_{23}(V_{\mu_{23}}) \,\oplus\, \Phi_h(V_{\mu_h}).$$

Proof. (i) Injectivity and T_i-equivariance for $i \le n-3$ are standard (May-20, Lemma 3.2; May-14 seventh pass, Lemma 3.2): for $i \le n-3$, the swap s_i moves only entries $i, i+1 \le n-2$, both inside the "inner" cells of μ_X (or μ_h); Hoefsmit coefficients depend only on inner cells, so T_i commutes with the corresponding Φ.

(ii) Pairwise disjointness of supports: each Φ_X's image is supported on SYTs of λ where $\{n-1, n\}$ occupies a specific unordered pair of cells:

- Φ_{12}: $\{(1,4),(2,3)\}$.
- Φ_{13}: $\{(1,4),(q+2,1)\}$.
- Φ_{23}: $\{(2,3),(q+2,1)\}$.
- Φ_h: $\{(2,2),(2,3)\}$ (with $n-1, n$ specifically at $(2,2),(2,3)$).

The four unordered cell pairs are pairwise distinct, so the supports are pairwise disjoint.

Dimension: $\dim \Phi_X(V_{\mu_X}) = f^{\mu_X}$ (injective), and $\dim E^+_{n-1}|_{V_\lambda}$ equals the number of SYTs of λ where $n-1, n$ are at the same row or at a removable corner pair. A direct enumeration over the four cell-pair configurations above gives $\dim E^+_{n-1} = f^{\mu_{12}} + f^{\mu_{13}} + f^{\mu_{23}} + f^{\mu_h}$, matching the summed Phi-image dimensions. By Theorem 8, $R'_n V_\lambda = E^+_{n-1}|_{V_\lambda}$, so the direct-sum decomposition follows.

[Computationally verified at $q \in \{2,3,4,5\}$: $\dim E^+_{n-1} V_\lambda$, $\dim R'_n V_\lambda$, f^{μ_X} for $X \in \{12, 13, 23, h\}$ all agree as claimed — see `~/projects/scratch/2026-05-14-extended-sign-kill-43/verify_decomp.py`.]

□ □

Remark 11. The $r = 2$ multi-corner shape $(4,3,1^q)$ has *three* corner pairs (so three 2-block branches), in contrast to the $r = 1$ shape $(3,3,1^q)$ of the seventh pass (only one corner pair). Among the three same-row candidates, only c_2 contributes (cf. the analogous analysis in the seventh-pass paper, where the same-row at $c_1 = (2,3)$ also contributed). The reason the same-row at $c_1 = (1,4)$ here is forbidden is the row-2 column-3 constraint: in $\widehat{\lambda} = (4,3)$, cell $(2,3)$ exists (since $\lambda_2 = 3$), whereas in $\widehat{\lambda} = (4,2)$ (May-14 sixth pass) cell $(2,3)$ does not exist and the analogous same-row at c_1 does contribute (Remark 5).

4 The two-branch reduction

Theorem 12 (Two-branch reduction)**.** *For $\lambda = (4,3,1^q)$ with $q \ge 3$,*

$$B \;=\; (T_{n-5}{+}1)(T_{n-4}{+}1)(T_{n-3}{+}1)(T_{n-2}{+}1)\Big[\,\Phi_{13}\big(B^{(\mu_{13})}_{\ell(\mu_{13})+1} V_{\mu_{13}}\big) + \Phi_{23}\big(B^{(\mu_{23})}_{\ell(\mu_{23})+1} V_{\mu_{23}}\big)\,\Big]. \qquad (1)$$

Proof. We follow the May-15/May-20 pull-through, applied separately to each of the four pieces of $R'_n V_\lambda$ from Lemma 10, then show two pieces vanish.

Step 1 (pulling four R'-blocks through Φ). Let $\Phi \in \{\Phi_{12}, \Phi_{13}, \Phi_{23}, \Phi_h\}$ and μ the corresponding inner shape.

Decompose $R'_{n-1} = (T_{n-2}+1)\,U_1$ where $U_1 = (T_{n-3}+1)\cdots(T_1+1) = R'^{(\mu)}_{n-2} \in H_q(S_{n-2})$. Since $T_1, \ldots, T_{n-3}$ commute with T_{n-1} (index gaps ≥ 2), U_1 preserves $E^+_{n-1}|_{V_\lambda}$. By Lemma 10(i), $U_1\Phi(v) = \Phi(R'^{(\mu)}_{n-2} v)$ for $v \in V_\mu$. Hence

$$R'_{n-1}\Phi(V_\mu) \;=\; (T_{n-2}+1)\,\Phi\big(R'^{(\mu)}_{n-2} V_\mu\big).$$

Iterating the same decomposition for $R'_{n-2}, R'_{n-3}, R'_{n-4}$, using that each U_k commutes with the previously-pulled T_j factors (index gaps ≥ 2):

$$R'_{n-4}R'_{n-3}R'_{n-2}R'_{n-1}\Phi(V_\mu) \;=\; (T_{n-5}+1)(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)\,\Phi\big(Q_\mu V_\mu\big), \qquad (2)$$

where

$$Q_\mu \;:=\; {R'}^{(\mu)}_{n-5}\,{R'}^{(\mu)}_{n-4}\,{R'}^{(\mu)}_{n-3}\,{R'}^{(\mu)}_{n-2}.$$

Step 2 (combining and applying Phase A). By Lemma 10(ii) and Theorem 8,

$$\begin{aligned} B \;=\; \Omega V_\lambda \;&=\; {R'}_{n-4}{R'}_{n-3}{R'}_{n-2}{R'}_{n-1}{R'}_{n}V_\lambda \\ &=\; {R'}_{n-4}{R'}_{n-3}{R'}_{n-2}{R'}_{n-1} \bigoplus_{X\in\{12,13,23,h\}} \Phi_X(V_{\mu_X}) \\ &=\; (T_{n-5}{+}1)(T_{n-4}{+}1)(T_{n-3}{+}1)(T_{n-2}{+}1)\,\sum_X \Phi_X\big(I_X\big), \end{aligned}$$

where $I_X := Q_{\mu_X}V_{\mu_X}$.

Step 3 (the μ_{12} piece vanishes). $\mu_{12} = (3,2,1^q)$ has $\ell(\mu_{12}) = q+2 = n-5$, hence $\ell(\mu_{12})+1 = n-4$. The inner quadruple product is

$$I_{12} \;=\; {R'}^{(\mu_{12})}_{n-5}\,\big({R'}^{(\mu_{12})}_{n-4}\,{R'}^{(\mu_{12})}_{n-3}\,{R'}^{(\mu_{12})}_{n-2}V_{\mu_{12}}\big) \;=\; {R'}^{(\mu_{12})}_{n-5}\,B^{(\mu_{12})}_{n-4}V_{\mu_{12}}.$$

By Theorem 3 applied to $\mu_{12} = (3,2,1^q)$ with $q' = q \geq 3$ (joint hypothesis of the theorem and the present proof; we will tighten to $q \geq 5$ below), $B^{(\mu_{12})}_{n-4}V_{\mu_{12}} \subseteq E_1^-|_{V_{\mu_{12}}}$. But ${R'}^{(\mu_{12})}_{n-5} = (T_{n-6}+1)(T_{n-7}+1)\cdots(T_1+1)$ has (T_1+1) as its *rightmost* factor, which annihilates $E_1^-|_{V_{\mu_{12}}}$. Hence $I_{12} = 0$ and the μ_{12} contribution drops out.

Step 4 (the hook piece vanishes). $\mu_h = (4,1^{q+1})$ has $\ell(\mu_h) = q+2 = n-5$, hence $\ell(\mu_h)+1 = n-4$. Same argument as Step 3:

$$I_h \;=\; {R'}^{(\mu_h)}_{n-5}\,B^{(\mu_h)}_{n-4}V_{\mu_h}.$$

By Theorem 7 applied to $\mu_h = (4,1^{q+1})$ with $k=4$, $m = q+1$ (hypothesis $m \geq \max(k,2) = 4$, i.e., $q \geq 3$ ✓), $B^{(\mu_h)}_{n-4}V_{\mu_h} \subseteq E_1^-|_{V_{\mu_h}}$. The rightmost (T_1+1) of ${R'}^{(\mu_h)}_{n-5}$ kills this, so $I_h = 0$.

Step 5 (the μ_{13} and μ_{23} pieces are standard $B^{(\mu_X)}$). Both $\mu_{13} = (3,3,1^{q-1})$ and $\mu_{23} = (4,2,1^{q-1})$ have $\ell(\mu_X) = q+1 = n-6$, hence $\ell(\mu_X)+1 = n-5$. The inner quadruple product runs from ${R'}^{(\mu_X)}_{n-5}$ to ${R'}^{(\mu_X)}_{n-2}$, exactly the four blocks of $\Omega^{(\mu_X)}$. Therefore

$$I_{13} \;=\; B^{(\mu_{13})}_{\ell(\mu_{13})+1}V_{\mu_{13}}, \qquad I_{23} \;=\; B^{(\mu_{23})}_{\ell(\mu_{23})+1}V_{\mu_{23}}.$$

Combining Steps 2–5 gives (1). □ □

Remark 13. The dimensions match nicely:

$$\dim B \;=\; f^{(3,2)} \;=\; 5, \qquad \dim B^{(\mu_{13})}_{\ell+1} \;=\; f^{(2,2)} \;=\; 2, \qquad \dim B^{(\mu_{23})}_{\ell+1} \;=\; f^{(3,1)} \;=\; 3,$$

and $5 = 2+3$ confirms that both contributing Phi-pieces inject and the four outer (T_j+1) factors are injective on the image (computationally verified at $q=5$).

5 Main theorem

Theorem 14 (Main)**.** *For* $\lambda = (4,3,1^q) \vdash n = q+7$ *with* $q \geq 5$,

$$B \;\subseteq\; \bigcap_{j=1}^{q-4} E_j^- |_{V_\lambda} \, .$$

Proof. By Remark 2, it suffices to show that every $T \in \operatorname{supp}(B)$ satisfies $p(T) \geq q-3 = n-10$. We use the two-branch reduction (1) together with prefix bounds on the two inner $B_{\ell+1}^{(\mu_X)}$ pieces and Lemma 9 on the four outer factors.

Step 1 (prefix bound on the Φ_{13}-image). $\mu_{13} = (3,3,1^{q-1})$ has size $n-2$, $\ell(\mu_{13}) = q+1$, and trailing-1-count $q' = q-1$. Since $q \geq 5$ we have $q' \geq 4$, so Theorem 6 applies, giving

$$p(T') \;\geq\; q'-2 \;=\; q-3 \qquad \text{for every } T' \in \operatorname{supp}(B_{\ell(\mu_{13})+1}^{(\mu_{13})} V_{\mu_{13}}).$$

Now $\Phi_{13}(v_{T'})$ is supported on the two SYTs of λ obtained from T' by placing $\{n-1,n\}$ at $\{c_1,c_3\} = \{(1,4),(q+2,1)\}$ in the two orders. Column-1 cells of μ_{13} are at rows $1,2,\ldots,q+1$ (i.e., $\ell(\mu_{13})$ rows). The added cells $(1,4)$ (column 4, not 1) and $(q+2,1)$ (row $q+2$ at column 1, with value $n-1$ or $n \neq q+2$) do not extend the identity prefix. Since $p(T') \leq q+1 = \ell(\mu_{13})$ and rows $1,\ldots,p(T')$ of column 1 are unchanged, we get $p(\Phi_{13}\text{-supp}) \geq p(T') \geq q-3$.

Step 2 (prefix bound on the Φ_{23}-image). $\mu_{23} = (4,2,1^{q-1})$ has size $n-2$, $\ell(\mu_{23}) = q+1$, and trailing-1-count $q' = q-1$. Since $q \geq 5$ we have $q' \geq 4$, so Theorem 4 applies, giving

$$p(T') \;\geq\; q'-2 \;=\; q-3 \qquad \text{for every } T' \in \operatorname{supp}(B_{\ell(\mu_{23})+1}^{(\mu_{23})} V_{\mu_{23}}).$$

The Phi-embedding Φ_{23} adds $\{n-1,n\}$ at $\{c_2,c_3\} = \{(2,3),(q+2,1)\}$: $(2,3)$ is in column 3 (not 1), and $(q+2,1)$ is in row $q+2$ with value $n-1$ or $n \neq q+2$. By the same argument as Step 1, $p(\Phi_{23}\text{-supp}) \geq p(T') \geq q-3$.

Step 3 (outer factors preserve the prefix). The four outer factors in (1) are $(T_j{+}1)$ for $j \in \{n-5,n-4,n-3,n-2\}$. Each index satisfies $j \geq n-5 = q+2 > q-2 = (q-3)+1$, so the hypothesis $j \geq P+1$ of Lemma 9 (with $P = q-3$) is met. Applying the lemma four times in sequence, the prefix bound $\geq q-3$ is preserved.

Conclusion. By Steps 1–3 and (1), every $T \in \operatorname{supp}(B)$ has $p(T) \geq q-3$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq q-4 = \tau(\lambda)$. □ □

Remark 15 (Hypotheses)**.** The hypothesis $q \geq 5$ is the joint requirement of:

- Theorem 6 at $q' = q-1 \geq 4$, i.e., $q \geq 5$ (for μ_{13}).
- Theorem 4 at $q' = q-1 \geq 4$, i.e., $q \geq 5$ (for μ_{23}).
- Theorem 3 at $q' = q \geq 3$, i.e., $q \geq 3$ (for μ_{12} vanishing in Step 3 of Theorem 12).
- Theorem 7 at $m = q+1 \geq 4$, i.e., $q \geq 3$ (for μ_h vanishing in Step 4).

The threshold for the two recursive-input theorems ($q \geq 5$) is tight: at $q = 4$ the prediction $\tau = q-4 = 0$ is vacuous, and at $q \leq 3$ the prediction is negative. Thus the hypothesis $q \geq 5$ exactly matches the first q at which $\tau \geq 1$.

6 Computational verification

We verify Theorem 14 and the structural decomposition computationally at $q = 5$ (so $n = 12$, $\dim V_\lambda = 1232$).

Decomposition. At $q = 5$,

$$f^{\mu_{12}} = f^{(3,2,1^5)} = 160, \qquad f^{\mu_{13}} = f^{(3,3,1^4)} = 225,$$
$$f^{\mu_{23}} = f^{(4,2,1^4)} = 350, \qquad f^{\mu_h} = f^{(4,1^6)} = 84.$$

Sum $= 819 = \dim R'_n V_\lambda = \dim E^+_{n-1}|_{V_\lambda}$, consistent with Lemma 10.

Vanishing. The two pulled-through inner pieces $\Phi_{12}(I_{12})$ and $\Phi_h(I_h)$ contribute 0 to B, leaving the two pieces $\Phi_{13}(I_{13}) + \Phi_{23}(I_{23})$ with $\dim I_{13} = 2$, $\dim I_{23} = 3$, summing to $5 = \dim B$.

E_1^- containment at $q = 5$. Apply Ω to five independent random vectors in V_λ (mod $P = 100\,003$ at $q = 11$) and check $(T_1 + 1)\,\Omega v = 0$ on each. All five tests pass. Probing $\dim B$ via Ω applied to the first 20 basis vectors gives rank 5, matching $f^{(3,2)} = 5$.

Sharpness of the $q \geq 5$ hypothesis. At $q = 4$ (where $\tau = q - 4 = 0$ is vacuous), the same random-vector test returns 164 nonzero entries in $(T_1 + 1)\,\Omega v$ on each of 5 random vectors — i.e., $B \not\subseteq E_1^-$ at $q = 4$. The boundary $q \geq 5$ for $B \subseteq E_1^-$ in the $\widehat{\lambda} = (4, 3)$ family is therefore sharp.

Scripts: `~/projects/scratch/2026-05-14-extended-sign-kill-43/verify_decomp.py, verify_E1minus_`
`check_q4_sharp.py`.

7 Updated table of proved hat_lambda rows

Table 1: Status of the extended sign-kill conjecture by $\widehat{\lambda}$. The τ column lists the predicted threshold $\tau(\widehat{\lambda}) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$.

$\widehat{\lambda}$	predicted τ	status	paper
$(2,2)$	$q-1$	proved $q \geq 2$	May-14 fourth pass
$(3,2)$	$q-2$	proved $q \geq 3$	May-14 fifth pass
$(4,2)$	$q-3$	proved $q \geq 4$ (gap closable)	May-14 sixth pass
$(3,3)$	$q-3$	proved $q \geq 4$	May-14 seventh pass
$(4,3)$	$q-4$	**proved $q \geq 5$ (this paper)**	May-14 eighth pass

8 Discussion

8.1 The recursive ladder is consolidating

This paper completes the second row of the recursive-ladder table: every $\widehat{\lambda}$ with $\widehat{\lambda}_1 + \widehat{\lambda}_2 \leq 7$ is now proved. The next natural target is $\widehat{\lambda} \in \{(5,2), (4,4), (5,3)\}$, each with threshold $\tau = q - 4$ or $\tau = q - 5$.

8.2 Branch-vanishing as a structural feature

Four pieces of E^{+}_{n-1}, two vanish, two contribute — this is the first case where *two* branches drop out. The pattern: a branch μ_X vanishes precisely when $\ell(\mu_X)+1=n-4$ (one less than the standard "$\ell+1=n-5$" alignment of μ_{13},μ_{23}), giving an extra inner $R'^{(\mu_X)}_{\ell(\mu_X)}$ factor whose rightmost (T_1+1) kills $B^{(\mu_X)}_{\ell+1}V_{\mu_X}$ via either the inductive extended sign-kill (here, the $(3,2)$ theorem at μ_{12}) or the hook column-1 theorem (at μ_h).

This bookkeeping — "which μ_X have $\ell+1=n-5$ vs. $n-4$" — is shape-uniform once the corner-pair indexing is fixed, and should generalize cleanly to deeper $\widehat{\lambda}$.

8.3 Sharpness

The sharpness direction ($B\not\subseteq E^{-}_{q-3}$ for $q\geq 5$) would follow a structural reduction analogous to the seventh-pass Sharpness Lemma (locating entry $n-7$ in a witness SYT and arguing via off-diagonal block + universal $T_\ell=q$). We leave the sharpness write-up for a follow-up note; the predicted threshold matches the computational data at $q=5$ (Table 1 of the conjecture paper extended to $q=5$: $B\not\subseteq E^{-}_{2}$ verified).

8.4 Push status

69 unpushed commits on `clio-vega/proofs` (after today's commit). PAT issue from May 6 persists — please refresh.